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COM3502-4502-6502
SPEECH PROCESSING

Lecture 13
Sounds and Symbols

COM3502-4502-6502 Speech Processing: Lecture 13, slide 1

1

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Writing Systems

Writing
Writing
Writing
Writing
Writing
Writing
Writing
Writing

- Writing is an ancient technical artefact that has been invented by people for use as a ‘communication aid’
- Writing allows information to be:
 - transmitted over space
 - stored over time
- The modern-day equivalent would be:
 - telephone (*voice transmission*)
 - voicemail (*voice store & forward*)
- Most languages have a writing system

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2

Writing Systems

Writing

Writing

Writing

Writing

Writing

Writing

Writing

Writing

- The basic unit of speech in most languages is the ‘word’ (because it carries meaning)
- Meaningful ‘utterances’ are made up from sequences of words
- Languages depict words using
 - alphabets, syllabaries, logographs or ideographs
- Writing systems that use alphabets and syllabaries are ‘phonetic’, i.e. they are based on how the words *sound*
- There is a closer relationship between **spelling** and **pronunciation**, in some languages (e.g. *Korean*) than in others (e.g. *English*)



العربي	български	català	中国话	hrvatski	česky
english	ελληνικά	עברית	हिंदी	italiano	日本語
한국어	românește	русский	српски	தமிழ்	

Question ...

How many sound segments in the word “six” ?

Answer = 4 (... not 3 !)

“six” → “siks”

The Syllable

- The 'syllable' is the shortest stretch of speech
- Syllables consist of ...
 - 'vowels': *sound segments produced using an unobstructed configuration of the vocal tract*
 - 'consonants': *sound segments in which the airflow is at least partly obstructed*
- A simple 'CVC' (consonant-vowel-consonant) syllable corresponds to the opening and closing of the mouth
- Words can be ...
 - monosyllabic (*having one syllable*)
 - polysyllabic (*having two or more syllables*)

Consonants and Vowels

Words can be described in terms of their 'CV-skeleton' ...

"eye"	→	V
"my"	→	CV
"eat"	→	VC
"book"	→	CVC
"draw"	→	CCV
"sign"	→	CVC
"signal"	→	CVCCVC
"strengths"	→	CCCVCCCC

Consonants and Vowels

Some more examples ...

"speech"	→	CCVC
"me"	→	CV
"meat"	→	CVC
"small"	→	CCVC
"open"	→	VCVC
"crisps"	→	CCVCCC
"exercise"	→	VCCVCVC or VCCVCCVC

Sound-Based Writing Systems

Omniglot a guide to written language

<http://www.omniglot.com/>

ᏍᏏᏉᏍᏏᏉ
עברית

- Syllabaries
 - symbols represent a consonant plus a vowel or a single vowel
 - examples: *Cherokee, Japanese*

Кирилица
Ελληνικά

- Abjads
 - symbols mainly represent consonants only
 - most abjads are written from right to left
 - examples: *Aramaic, Hebrew*

G W Y
カタカナ

- Alphabets
 - symbols represent consonants and vowels
 - examples: *Cyrillic, Greek*

IPA Consonants

CONSONANTS (PULMONIC)												
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal	
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ	
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ			
Trill	ʙ			r					ʀ			
Tap or Flap				ɾ		ɽ						
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ	
Lateral fricative				ɬ ɮ								
Approximant		ʋ		ɹ		ɻ	j	ɰ				
Lateral approximant				l		ɭ	ʎ	ʟ				

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

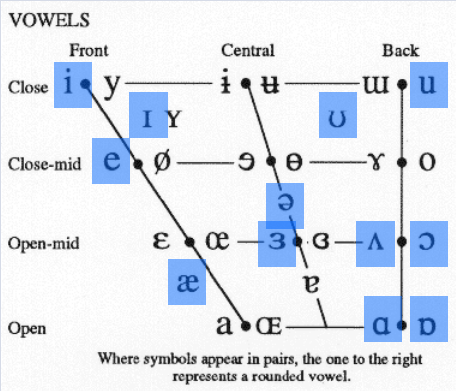
Cantonese Consonants

The English Consonants

p	<u>p</u> ea	z	<u>z</u> oo
t	<u>t</u> ea	ʒ	a <u>z</u> ure
k	<u>k</u> ey	tʃ	<u>ch</u> in
b	<u>b</u> ay	dʒ	<u>j</u> am
d	<u>d</u> ay	m	<u>m</u> ap
g	<u>g</u> ay	n	<u>n</u> ap
f	<u>f</u> in	ŋ	han <u>g</u>
θ	<u>th</u> in	ɹ	<u>r</u> ed
s	<u>s</u> ip	l	<u>l</u> ed
ʃ	<u>sh</u> ip	w	<u>w</u> et
v	<u>v</u> at	j	<u>y</u> et
ð	<u>th</u> at	h	<u>h</u> it

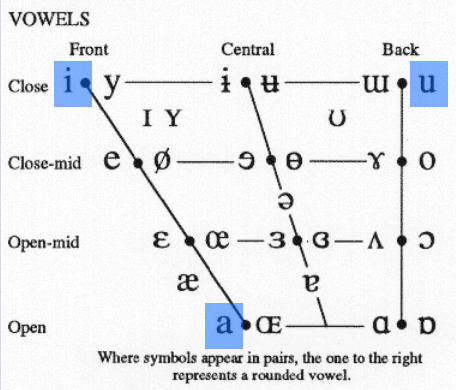
Videos of pronunciation:
<https://learnipa.group.shef.ac.uk/IPAChart/index.html>

IPA Vowels



English Vowels

IPA Vowels



Arabic Vowels

The English Vowels

Monophthongs	i:	h <u>ee</u> d	eɪ	h <u>ay</u>	Diphthongs
	ɪ	h <u>i</u> d	aɪ	h <u>igh</u>	
	e	h <u>ea</u> d	ɔɪ	b <u>oy</u>	
	æ	h <u>a</u> d	əʊ	h <u>oe</u>	
	ɑ:	h <u>a</u> rd	aʊ	h <u>ow</u>	
	ɔ:	h <u>oa</u> rd	ɪə	h <u>ear</u>	
	ɒ	h <u>o</u> d	eə	th <u>ere</u>	
	ʌ	h <u>u</u> d	ʊə	m <u>oor</u>	
	ʊ	h <u>oo</u> d			
	u:	wh <u>o</u> 'd			
	ɜ:	h <u>ear</u> d			
	ə	<u>a</u> bout			

23

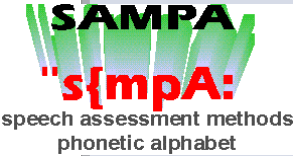
Some Examples

“speech”	→	[spi:tʃ]
“me”	→	[mi:]
“meat”	→	[mi:t]
“small”	→	[smɔ:l]
“open”	→	[əʊpən]
“crisps”	→	[kɹɪsps]
“exercise”	→	[eksəsaɪz] or [eksəɪsaɪz]
“strengths”	→	[stɹɛŋkθs]

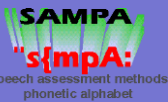
24

Machine Readable Phonetic Alphabets

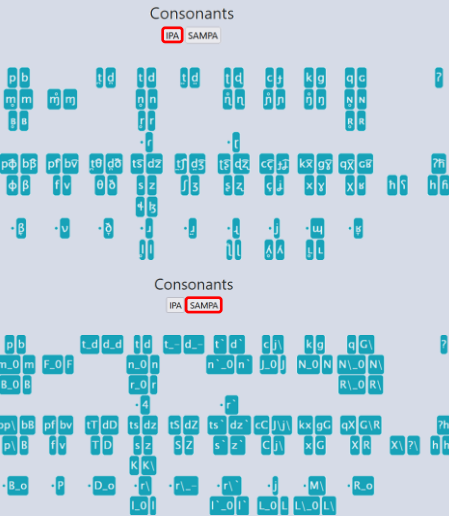
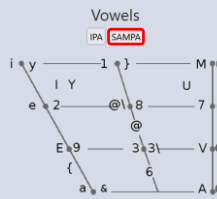
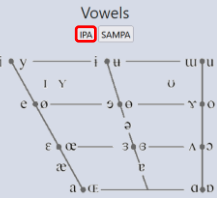
Symbol	Orthography	SAMPA	IPA
...	...	...	...
f	<u>f</u> in	fIn	fin
T	<u>th</u> in	TIn	θin
v	<u>v</u> im	vIm	vim
D	<u>th</u> is	DIs	ðis
s	<u>s</u> in	sIn	sin
S	<u>sh</u> in	SIn	ʃin
z	<u>z</u> ing	zIN	ziŋ
Z	me <u>as</u> ure	meZ@	meʒə
...	...	...	...



<http://www.phon.ucl.ac.uk/home/sampa/>



SAMPA vs. IPA



https://www.lfsag.unito.it/ipa/index_en.html



Finally ...

Why don't all languages adopt a phonetic-based writing system?

... because it would be much harder to understand the writing of people who speak with different accents or dialects

This is why a non-phonetic system (such as *Chinese logograms*) is beneficial for a large population

This slide set has covered ...

- Writing systems
- The syllable
- Consonants and vowels
- Phonetic transcription
- The 'International Phonetic Alphabet'
- Machine-readable phonetic alphabets

Any Questions ?



36

Next slide set ...

Articulatory Phonetics

37